SIMATS Engineering CO2-AT3- Assesment Class-test	NAME: G. Joel Richard Reg No: 192524084 COURSE CODE : DSA0204 COURSE NAME : computer vision with opencv

1. An image appears too bright. Explain and Justify the most suitable grey level transformation to improve visibility.

Ans: The most suitable grey level transformation is negative transformation. It converts bright pixels into dark pixels, redusing excessive brightness. This improves image visibility.

2. A dark image lacks detail. Describe and justify an appropriate grey level transformation to enchance visibility.

Ans: An appropriate transformation is log transformation. It increases the intensity of dark pixels while keeping brighter pixels while keeping brighter pixels almost unchanged.

3. An image has poor contrast. Explain a suitable histogram-based technique for contrast enchancement.

Ans: The suitable technique is histogram Equalization. It redistributes pixel intensity values over a wider range, increasing image contrast. This makes objects and image features more visible.

6. A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter.

Ans: The mean filter is suitable for reducing random noise. It replaces each pixel with the average value of neighboring pixels.

8. An image appears blurred after smoothing. Explaind and justify a suitable sharpening method to restore details.

Ans: The Laplacian sharpening method is used restore details. It enhances high-frequency components such as edges, making the image appear sharper.

15. An object is clearly brighter than the background Explain and justify the segmentation technique used.

The suitable technique is Thresholding. A threshold value is selected, and pixels above the threshold are classified as the object while the remaining pixels become the background. This provides simple and effective segmentation.

17. An image requires identification of object edges. Explain and justify a suitable edge detection method.

The global edge detection method is used. It detects intensity changes in horizontal and vertical directions to identify object boundaries accurately.

20. You need to extract object outlines from an image. Describe and apply an appropriate technique.

use boundary detection. It identifies the outer edges of segmented objects and extracts their outlines. this is useful for object recognition and shape analysis.

23. An image contains salt and pepper noise. Explain and Justify the most appropriate spatial filter.

Ans: The median filter is the best choise. It replaces each pixel with the median value of neighbouring pixels. It effectively removes salt and pepper noise while preserving image edges.

27. A grayscale image shows poor seperation between object and background intensities. Describe and justify a thresholding approach.

Ans: use global threshoulding. A single threashold value is applied to the entire image to seperate the object from the background. It is simple, fast, and works well when there is a suffiient contrast between the object and the background.